DECODELABS · INTERNSHIP SUBMISSION

# Iris Flower Classification Using AI

An end-to-end supervised learning pipeline built as an AI Engineer project.

**K-Nearest Neighbors · StandardScaler · Python**

**Verified test accuracy: 93.33%**



PROJECT 02

## Data Classification

2026 · Portfolio presentation

# Problem & Objective

THE QUESTION

## Can a machine learn to identify an Iris species from flower measurements?

Input: 4 numerical measurements
Output: 1 of 3 species classes

PROJECT OBJECTIVE

- Load and understand a labeled dataset
- Split data into training and testing sets
- Apply a simple supervised algorithm
- Validate the model with multiple metrics

**Outcome: a reproducible, reviewable ML pipeline.**

# Dataset: Iris Benchmark

**150**
SAMPLES

**4**
FEATURES

**3**
CLASSES

**0**
MISSING VALUES

## FEATURES USED

| | |
|---|---|
| Sepal length | cm |
| Sepal width | cm |
| Petal length | cm |
| Petal width | cm |

## CLASS BALANCE




Balanced classes support fairer evaluation.

# From Raw Data to Predictions

01

## Load

Iris dataset

→

02

## Split

80 / 20 stratified

→

03

## Scale

StandardScaler

→

04

## Train

KNN, k=5

→

05

## Evaluate

Metrics + matrix

**Design principle: keep preprocessing inside the model pipeline so the same transformation is applied consistently at training and inference time.**

# Scaled K-Nearest Neighbors

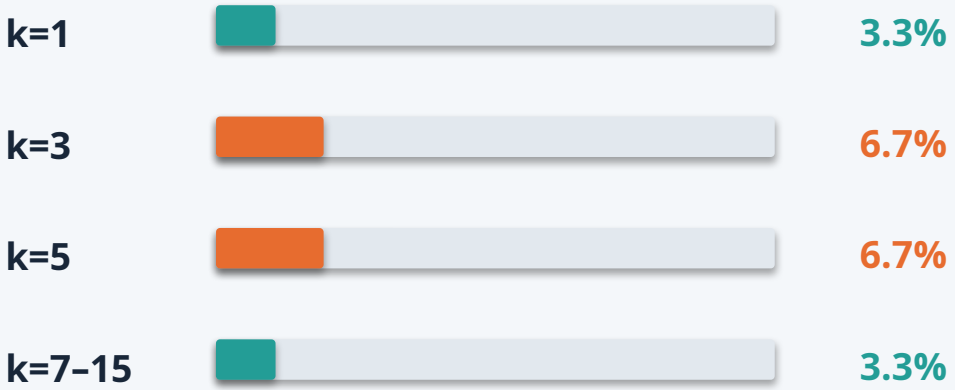
WHY KNN?

## Similar flowers tend to occupy nearby points in feature space.

- Simple and interpretable
- Strong fit for a small numerical dataset
- Scaling prevents large-unit features from dominating

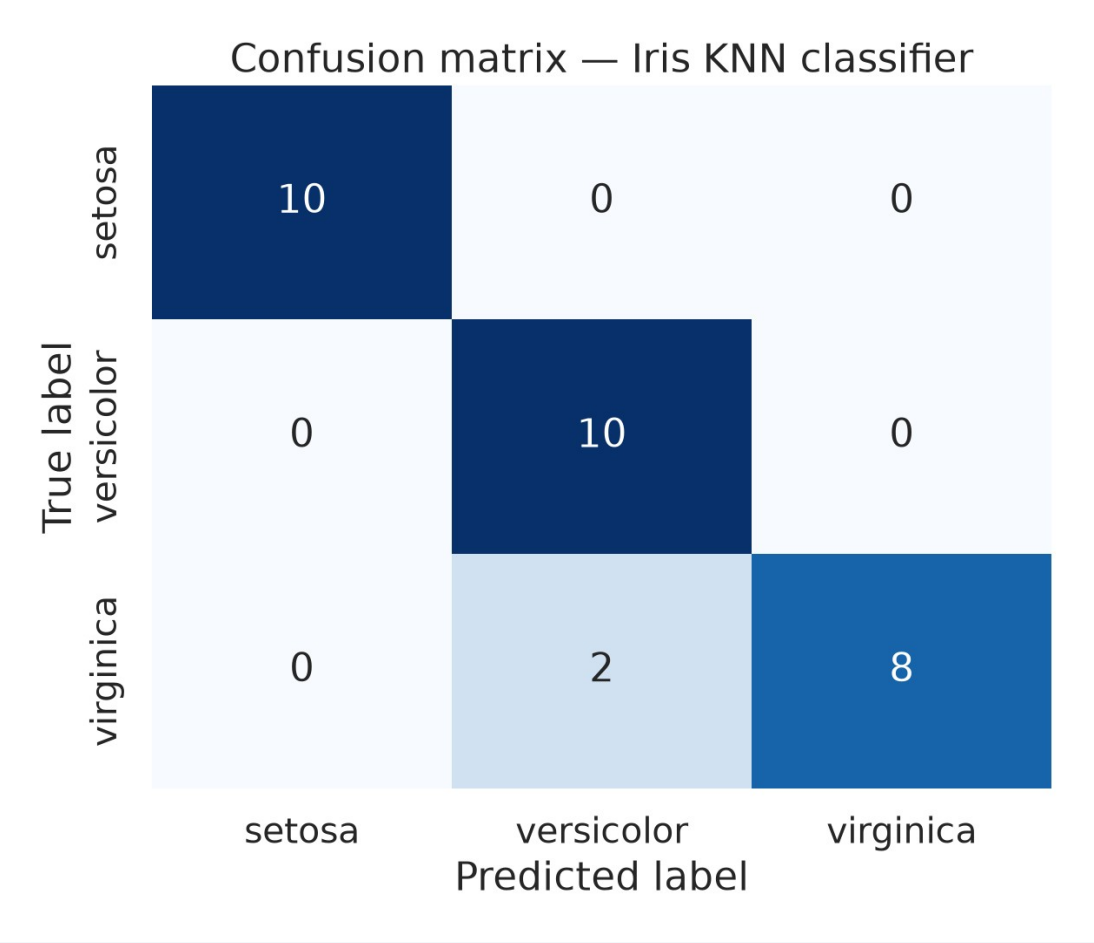
TUNING K

## The project tests odd k values from 1 to 15.




**Final configuration: StandardScaler → KNN(k=5)**

# Evaluation: What Did the Model Miss?




READING THE MATRIX

## Setosa was classified perfectly.

The only errors were two Virginica flowers predicted as Versicolor. This is the expected harder boundary between these visually similar classes.

**28 / 30 test samples correct**

Source: executed project artifact · confusion_matrix.png

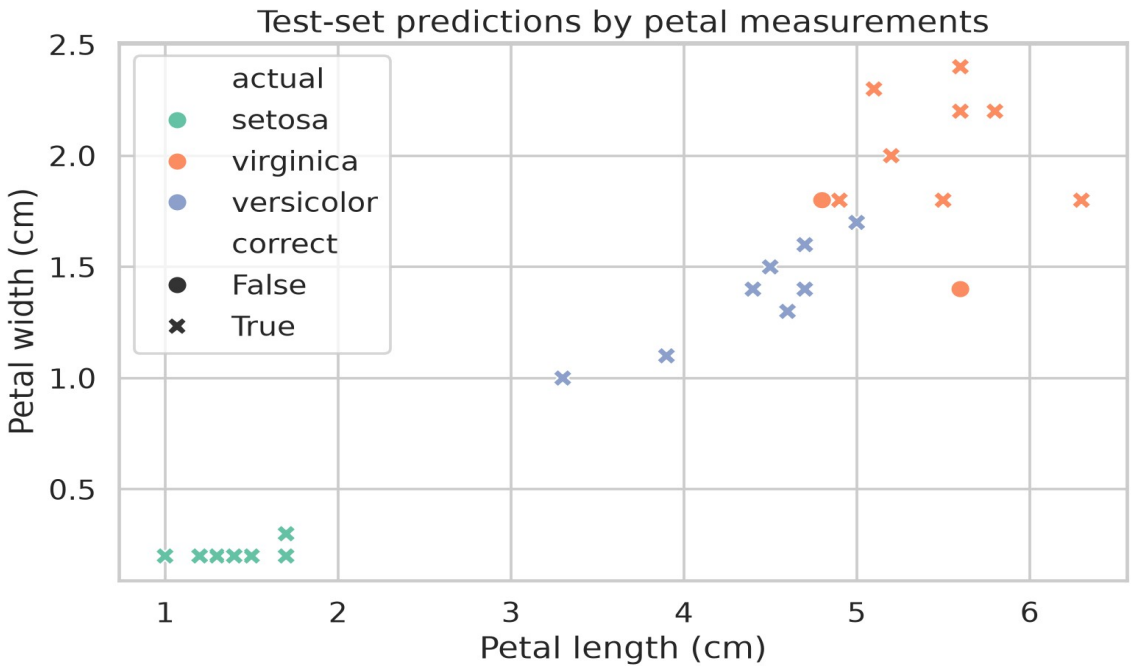
# Results & Evidence

**93.33%**
ACCURACY

**94.44%**
PRECISION

**93.33%**
RECALL

**93.27%**
WEIGHTED F1




**WHAT THIS DEMONSTRATES**

- Reproducible data preparation
- Distance-aware model selection
- Multi-metric model validation
- Clear error interpretation

Source: executed project artifacts · metrics.json and prediction_scatter.png

07 · CLOSE

# A small model, built like a professional system.

This project bridges theory and practice: data understanding, supervised learning, reproducible evaluation, and communication of results.

**NEXT PRODUCTION STEPS**

- Cross-validation and experiment tracking
- API deployment and monitoring
- Data drift and model governance

## Thank you

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